Supplementary Materials (SM):

SCREEN-LCA: Ex-ante prioritization of bio-based pathways for chemicals and plastics under a climate-mitigation objective

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Overview of the supplementary materials (SM)

Included in this document (mmc1.pdf):

- Supplementary information text, tables and references
- Connected supplementary files
- Kumu graph database reference details and links

Additional supplementary materials of the article:

- Step 2 - EXIOBASE results “mmc2.xlsx”
- Step 3 - PRODCOM results “mmc3.xlsx”
- Step 4 - Figure 2 data - Hotspot products “mmc4.xlsx”
- Git repository with Python notebooks and data (see “Git repository” section below)
- Graph database of hotspot products and their bio-based pathways: https://kumu.io/nicolas-lienart/screen-lca-bio-based-pathways-database

The main outputs are summarized in the section **Error! Reference source not found.** “**Error! Reference source not found.**” at the end of this file.

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Abbreviations

- GWP: Global warming potential, generally evaluated on a 100 years range (GWP_{100}), excluding the biogenic CO_2.
- GHG: Greenhouse gas
- LCA: Life cycle assessment
- PA: Polyamide
- FR: France
- SM: Supplementary materials
- EU27: European Union and its 27 member states as of 2020
- HS22: Harmonized System nomenclature 2022 edition

1. Git repository

In an effort to make this research reproducible, especially for the results of the illustrative demonstration, and to promote the reusability of the SCREEN-LCA method, all of the Python code and some of the generated data are published with an open licence under the Creative Commons public licence CC-BY 4.0.

The code and data are hosted on a Git repository, available on two platforms:

- Git repository (Forge INRAE): https://forge.inrae.fr/nicolas.lienart/screen-lca-paper-supplementary-code
- Git repository (GitHub): https://github.com/nicolnt/screen-lca-paper-supplementary-code

The Git repository is organized as a set of well-documented Jupyter-based interactive notebooks which are also visible online in read-only mode. It also includes a main "README.md" file at the root which provides general instructions, context and links, it also points toward the relevant Python notebooks. Each of these notebooks contains a "README" section at the top which describe the file and how to use it, along with some metadata.

Some input, intermediate and output data is also available in the repository. There are some exceptions due to licensing or size constraints (e.g., raw EXIOBASE, PRODCOM, or Ecoinvent data). In either case, the links to the relevant datasets are listed with their explicit version.

This repository and its content may receive future updates after journal publication. To access the repository and its files at a state matching exactly with the one used for the published version of the article, the Git tag "v1.0", standing for "Published version 1.0", can be used:

- Direct GitHub link: https://github.com/nicolnt/screen-lca-paper-supplementary-code/tree/v1.0
- Direct INRAE Forge link: https://forge.inrae.fr/nicolas.lienart/screen-lca-paper-supplementary-code/-/tree/v1.0

2. EXIOBASE (step 2)

2.1. EXIOBASE version and setup

EXIOBASE IOT version 3.10.2 is used in the product-by-product format ("*pxp*"), allowing for a greater level of details as compared to industry-by-industry ("*ixi*"). Although time series are available for up to 2024 in EXIOBASE versions 3.10.x, air emissions values, which include GHG emissions are now-casted from 2022 using aggregate statistics (Stadler *et al.*, 2026). In consequence, these more recent years of EXIOBASE data may lack representativeness at the product scale and should therefore be used with caution (Rasul, 2025, personal communication). For this reason, we use the year 2022, considered as the most representative.

As of the time of the writing, EXIOBASE versions 3.10.x onward do not provide characterized accounts, which were formerly known as "impact". It only comes with "stressors" representing uncharacterized emissions. Characterized impacts are planned for EXIOBASE 3.11 (Rasul, 2025 and 2026, personal communication). This implies that the user shall take the responsibility for characterization of EXIOBASE the stressors. Taking inspiration of the code of Aguilar-Hernandez (2025) and the "pymrio" Python library documentation[1], we build a characterization matrix

[1] https://pymrio.readthedocs.io/en/latest/notebooks/stressor_characterization.html

specifying the climate change GWP_{100} characterization factors of each GHG air emission accounted in EXIOBASE. We took the GWP_{100} factors from the IPCC AR6 method, extracting them via the "Brightway25" Python library. The procedure is further documented in the Python notebook for EXIOBASE calculation available in the Git repository.

2.2. Details on the EXIOBASE calculations

The GHG footprints of chemicals and plastic were estimated for both local production and imports for the specified region (i.e., France in the main article's demonstration). EXIOBASE datasets come with several matrices, with the most relevant for this method being described in Table S1.

Table S1: Structure and description of environmentally extended input-output matrices for demand and extension used in the case study.

Name	*Code*	*Dimensions (rows × columns)*	*Structure*	*Description*
Final demand	Y	Products by country/region × Products by country/region	Columns represent the final demand categories for each region. Rows correspond to the regions and products fulfilling the final demand categories.	Final households, governments and related consumptions, with capital formation (Eurostat, 2008). Final markets for their production, such as personal consumption purchases and sales to the federal government (Miller and Blair, 2022).
Intermediate demand	Z	Products by country/region × Products by country/region	Rows represent the production and columns the intermediate demand origin (region and product industry).	Consumption of industries. When a product is sold to a business, which could be in the same sectors, it is referred to as an inter-industry transaction (Miller and Blair, 2022).
Multiplier of the extensions	M	Impact indicators × Products by country/region	Rows represent the different impact categories covered by the IO extension. Columns correspond to the regions and products in the same order as the other IO matrices (e.g., "Y" matrix, "Z" matrix).	Total, direct and indirect, requirement factors for one unit of output (Stadler, 2021). Also called total impact multipliers (Wiedmann, 2017).

To align with the accounting approach defined in step 1 of the main article, four inputs from EXIOBASE were used to estimate the GHG footprint of the different sectors of a selected region (e.g., France, EU27). These inputs are as follows (Table S2):

- The first two consists of the footprints of production dedicated to intermediate (*a*) and of final demand production (*b*), which are calculated at the gate of French industries. This includes both the direct emissions and the embedded impacts of inputs, whether they come from other French industries or from imports
- The other two are the footprints of imported products dedicated to any intermediate (*c*) and of the final demand imports (*d*), calculated following the same logic as for the two first categories.

Table S2: Details on the initial computation procedures to obtain four matrices used in step 2 of the demonstration.

	Production	*Imports*
Consumers	• Intermediate demand (*a*) • Final demand (*b*)	• Intermediate demand (*c*) • Final demand (*d*)
Type of flow	Region's industries output product footprints.	Region's product imports footprints.
Description	Cradle-to-gate (not embedding impact of use and end of life).	Gate of the region, not embedding impact of last transportation step, use and end of life.
Procedure	Set values of rows outside of the region (i.e., all final demand origins except from the region) to 0.	1. Extract region's column values (i.e., region's demand). 2. Set region's row values (locally satisfied demand) to 0.

The instructions provided in Table S2 result in four matrices (*a*, *b*, *c* and *d*), which share a homogeneous row structure but have different columns. Subsequently, the following procedure was applied repeatedly for each of those matrices, hereafter denoted by the variable "A":

1. Sum columns of the matrix $A_{m \times n}$ to a vector $\mathbf{v}_{m \times 1}$. Resulting in a homogeneous column vector:
   $$\mathbf{v} = A1_n$$
2. Diagonalize $\mathbf{v}$ to square matrix $\hat{\mathbf{v}}$:
   $$\hat{\mathbf{v}} = diag(\mathbf{v})$$
3. Multiply the matrix containing the footprint impact coefficients ("M" matrix) with the resulting diagonal matrix $\hat{\mathbf{v}}$:
   $$R = M\hat{\mathbf{v}}$$
4. The columns are aggregated by product, summing values for the products associated to its different regions.

2.3.Total footprint (production and consumption) of a region

In the main article's Step 2, we compared the results to the total footprint of France and to the its total direct emissions. For consistency, we have derived these from EXIOBASE data, using the same version and year. Just like for calculating individual products footprint, stressors (e.g., GHGs) must be characterized, characterization is not provided in recent versions of EXIOBASE. Refer to section "Stressors characterization" of the EXIOBASE notebook for more details on the procedure used.

This step generates two helpful matrices:

- D_{pba} the total territorial emissions; and
- D_{cba} the total territorial footprint

They combine the direct emissions due to the final demand of the region (F_Y) to the direct emissions of the industry (F_s) and the footprint of all industry inputs to the final demand ($D_{cba,s}$) respectively. Total territorial footprint or emissions per region i are given by summing over the stressors (i.e., GHG) per sector ($0 \dots m$) plus the stressors directly linked to final consumption for all final demand categories ($0 \dots w$) of that region (Wood *et al.*, 2020; Stadler, 2021):

$$D_{pba}^{i} = \sum_{s=0}^{m} F_{s}^{i} + \sum_{c=0}^{w} F_{Yc}^{i}$$

$$D_{cba}^{i} = \sum_{s=0}^{m} D_{cba,s}^{i} + \sum_{c=0}^{w} F_{Yc}^{i}$$

2.4. Step 2 data and code

The four resulting vectors derived from the procedure described above, along with a summary of data, are provided in the Excel file "Step 2 - EXIOBASE results" (mmc2.xlsx) in the SM.

The following Python notebook in the Git can be used to reproduce the calculations: "Python_notebooks/Calculate sectors footprint with EXIOBASE.ipynb"

3. PRODCOM (step 3)

3.1. PRODCOM data gaps

At the country level, some products show data gaps. For example, there are several prominent chemicals which completely lacked data for France from 2013 up to 2024 (e.g., styrene, butene, propylene oxide, and adipic acid), 2024 being the latest year available at the time of the study. As illustrated, this does not only impact niche products and many commodities, such as the ones listed previously, are impacted. Therefore, taking "EU27_2020" dataset as the reference appears as a good alternative for the reason that this aggregated level the data is less prone to show data gaps, mostly due to confidentiality. With EU27 region's data, we saw much fewer products being fully impacted over the whole time range. France is a key player in the EU27 chemical sector (de Souza *et al.*, 2024).

3.2. Filtering PRODCOM products via the NACE classification

PRODCOM code is constructed based on a combination, from left to right, of NACE (first 4 digits), CPA code (digit 5 and 6), completed by a PRODCOM-specific code (rest). Each level adds more distinction between products than the parent level (Eurostat, 2023a).

The illustrative case study presented in the main article sets focus on pure petrochemicals and raw plastic polymers. This excludes inorganic substances and molecules that are already bio-based. In addition, this excludes formulations (e.g., consumer products or mixed chemicals) and composites materials (e.g., consumer products or industrial intermediates that already contain multiple materials). The PRODCOM products were filtered based on the NACE Rev 2.1 codes, permitting to directly filter out products falling outside the scope (EC, 2022; Eurostat, 2025).

The two following categories (20.14 and 20.16), corresponding to the scope, were selected in the database (Eurostat, 2023b):

- C MANUFACTURING (en)
    - 20 Manufacture of chemicals and chemical products (en)
        - 20.14 Other organic basic chemicals (en)
        - 20.16 Plastics in primary forms (en)

This was particularly efficient to filter out formulated products.

3.3. Excluded PRODCOM products

In the case study several products were considered too broad and excluded. For example, "Other oils and oil products, n.e.c." is the highest-ranking product by mass in the EU27 but excluded because it represents a heterogeneous mix rather than a specific product. These exclusions were based on three constraints:

1. Broad PRODCOM products can constitute chemically different compounds which may have distinct alternative pathways;
2. Each disaggregated component is likely to represent a smaller share which is artificially inflated by grouping;
3. Disaggregation requires additional data that is time-consuming to obtain and may not even be available.

Other exclusions are found in the Excel SM, where some products are categorized as follows:

- "Too broad": PRODCOM products representing a range of products that could not be disaggregated.
- "Bio-based": Products already bio-based.
- "Inorganic": Products that are not organic molecules.

3.4. Step 3 data and code

The Python notebook used to load, filter and format the data from PRODCOM is available in the Git repository at this location: "Python_notebooks/Extract PRODCOM data.ipynb"

The SM Excel file "Step 3 - PRODCOM results" (mmc3.xlsx) contains the extracted data which was manually filtered to remove out-of-scope products and subsequently mapped to Ecoinvent market activities and geographies (see the "Footprint estimations (step 4)" section below).

4. Footprint estimations (step 4)

4.1. Estimating total upstream footprint with Ecoinvent

In the illustrative example, the "upstream footprint" refers to the direct and indirect GHG emissions that occur due to the production and transport to consumer of the different products assessed during step 4. They are estimated by using the Ecoinvent database v3.12, the latest version available. We specifically selected "market for …" activities (i.e., unit processes) (Ecoinvent, 2025). These are meta-activities that are representative of the chosen geographical market in terms of process and production location. There exist multiple geographies (i.e., regions or "location" as per Brightway terms) for many of the Ecoinvent activities. We explain how we estimated the share of each geography using BACI in the section "Estimating suppliers' location with BACI" below. These market activities embed the impacts of transportation to the consuming market. The footprint is estimated

via the Brightway25 Python library, using the IPCC 2021 "climate change: total (excl. biogenic CO_2)" method based on the GWP_{100} indicator. For more details, refer to step 4 in the main article.

4.2. Estimating suppliers' location with BACI

Ecoinvent activities are often divided into multiple geographies[2]. This makes it possible to derive closer approximations of their real footprints based on regional specificities (e.g., electricity mix). It could theoretically be sufficient to use generic data which would not be representative of a very specific location of the process. However, Ecoinvent sparingly provide such a global ("GLO") geography, forcing the user to pick the "correct" representative geography of the production process. In addition, Ecoinvent's "market for …" datasets are often technically representative of production mix, not of a "consumption mix", contrary to what is stated in the database documentation[3]. Ecoinvent staff have confirmed this with us (Ecoinvent, 2026, personal communication). Taking either Europe ("RER") or Europe without Russia ("RER w/o RU") Ecoinvent's geographies does not necessarily make a good proxy to estimate the product footprints in the illustrative example context. Sometimes, a significant fraction of the PRODCOM region's available product quantity is imported and thus produced from outside Europe. It is the case for methanol, for example, of which 86% of the available quantity is imported, which is not reflected in Ecoinvent database as of v3.12.

In spite of the estimated share of local production and importations of each of the PRODCOM products, PRODCOM lacks the information of the supplying regions and their share. It is not sufficient to select the representative Ecoinvent geographies at this point. This is made more challenging as recent versions of Ecoinvent have further expanded the geographical granularity of some activities, thereby removing larger regions which would otherwise be suitable for serve as generic proxies.

To address this issue, we bring under play an additional database, CEPII's BACI, which allows us to derive the missing details regarding traded flows. BACI is introduced in the main manuscript's Table 2. It contains a country-level geographical coverage of trade data for around 5,000 products, using the 6-digit Harmonized System (HS) classification. BACI reconciliates UN Comtrade data, it provides both monetary (thousand USD) and quantity (tons) (Gaulier and Zignago, 2010). The version of BACI we used (i.e., V202501) is built on HS nomenclature 2022 edition (HS22). PRODCOM product are mapped to an HS22 code. It builds on top of the official correspondence between PRODCOM and the Combined Nomenclature (CN) provided by the EU[4]. CN is a derived classification compliant with the HS nomenclature[5].

For the article's demonstration, matching products to an HS22 code was straightforward, there were mostly one-to-one matches and only a few one-to-many which required user intervention to pick the most relevant. Once products were matched, we have made use of BACI to determine which countries or regions supply EU27 imports and their respective share.

For reproducibility and reuse, we provide a Python notebook containing the code to process the data. Its main functions are to obtain the list of providers of a given HS22 product to a given region

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[2] https://geography.ecoinvent.org/

[3] https://support.ecoinvent.org/glossary-m#M-Market-Activity and https://support.ecoinvent.org/market-activities

[4] https://showvoc.op.europa.eu/#/datasets/ESTAT_Combined_Nomenclature__2024__CN_2024/unknown/data

[5] https://taxation-customs.ec.europa.eu/customs/common-customs-tariff-cct/tariff-classification-goods/combined-nomenclature_en

(i.e., country or association, in our case the EU27) and calculate the share in physical quantity in coherence with PRODCOM's data. The code automatically conducts suppliers' aggregation at continental Europe level, excluding all the EU27 member states. For each product, the code calculates the the sum of extra-EU27 yet European supplier share. This share is subsequently combined with the local production share, thus yielding the effective share of production originating from Europe. This resulting share is mapped to Ecoinvent's "RER" geography. The same is done for Europe excluding Russia ("RER w/o RU"), a frequent geography since Ecoinvent's 3.12. The other external supplying country shares are mapped to the other relevant Ecoinvent geographies. Some Ecoinvent activities are only available for global ("GLO"), in such cases it will be affected 100% of the production. In other situations where "Rest-of-World (RoW)" geography is available, this RoW dataset handles the leftover calculated import shares which do not have a matching Ecoinvent geography. For example, the procedure estimated that 56% of ethylene glycol available in the EU27 comes from European producers ("RER"), the rest (44%) is mapped to RoW.

4.3.Selected substance contribution to the footprint

Most of the petrochemistry relies on a small subset of intermediate molecules, called building blocks or bulk chemicals, among which we mostly find ethylene, propylene, and a trio made of benzene, toluene and xylene isomers (BTX) (IEA, 2018; Petrochemicals Europe, 2019). We wanted to highlight their contribution to the different molecules footprint in order to see how relevant it could be to target these as part of a substitution strategy rather than the downstream products they integrate. For this part and in the Python code, we refer to them as "substances", to distinguish them from the hotspot products with the same name.

Automatic approach (general case)

We leverage the structure of Ecoinvent database, which by its matrix construction is built as a network interconnecting the activities with their inputs. To calculate the share of each of these substances we have implemented a recursive graph traversal function which starts from a hotspot product and searches for a given substance by gradually moving upstream of the production pathway until found or reaching a cutoff point (i.e., max node traversed and min absolute upstream product amount). Refer to the "`lca_contribution_of_substance`" function of the footprint calculation Python notebook.

Some Ecoinvent activities are aggregated, where inputs would be embedded therefore not connected to external activities. This is the case of Polyamide -6 and -6,6 which we handled differently as described in the section below. Otherwise, all other hotspot products in the top 15 list could leverage this automatic network-based function.

Specific situation of polyamide -6 and -6,6

Polyamide -6 (PA-6) and -6,6 (PA-6,6) intermediate chemical footprint contributions were estimated based on proxy because the specific Ecoinvent datasets for these products are aggregated (also referred to as system processes by SimaPro[6]). This implies that all the intermediates required directly by the process or intervening upstream are not public, instead an aggregated life cycle inventory is given (elementary flows) (UNEP, 2011). This demanded some manual work to plot the substance contribution of those two products in the Panel B of Figure 2 in the main article. We used some assumptions and literature references to estimate the contributions of the selected substances.

[6] https://support.simapro.com/s/article/What-are-unit-and-system-processes (accessed: 02/02/2026)

Polyamide -6 is obtained from ε-caprolactame ring opening polymerization (Herzog *et al.*, 2020; Khedr, 2023) where 1 kg is required per kg of polymer (Li, 2007). ε-caprolactame is not part of Ecoinvent database and we found no relevant proxy. Thus, we estimated the benzene requirements for caprolactame production using data from the literature, which corresponds to 0.95 kg of benzene per kg of ε-caprolactame (Hong and Xu, 2012). Again, benzene is available as a disaggregated dataset in Ecoinvent.

On the other hand, polyamide -6,6 (PA 66) is the result of a polycondensation reaction of adipic acid and hexamethylene diamine (HMDA) (Herzog *et al.*, 2020; Khedr, 2023). We used the data from the literature to estimate both HMDA and adipic acid intermediates requirements to respectively 0.5 kg/kg PA 66 and 0.65 kg/kg PA 66 (Li, 2007).

4.4. End-of-life footprint

The identified top products may have their own possible final disposal fates, which can be a represented by a combination of end-of-life scenario. These scenarios are bound the downstream chemicals or objects that the intermediates products will eventually compose. Moreover, the end of life can very greatly by region and sector (Geyer, Jambeck and Law, 2017; Ryberg, Laurent and Hauschild, 2018; OECD, 2022a). Unfortunately, due to a lack precise and reliable data, we could not map each product (i.e., chemical or polymer) to a specific representative end-of-life scenario. Attempting to track the life cycle path of an intermediate molecule from its production all the way down to the final disposal of the end-products is overwhelmingly data intensive, with great risk of errors. Not to mention the chemical reactions which sometimes split the molecules into several components.

Nonetheless, due to their fossil nature, skipping the end-of-life part in the assessment of the hotspot products' GHG footprints would be misleading. To go around the complexity of representing each product's single case specifically in this work we assume a common fate:

First, based on the literature, we assume that most of the top molecules in the list (e.g., ethylene, propylene, BTX), and their derivatives, are used in the plastics industry (IEA, 2018; Petrochemicals Europe, 2019; Meys *et al.*, 2021). Then, as we do not know if the produced and imported quantities are effectively disposed in the same region, we do not consider the end of life specifically for the region under study. Therefore, we took the global average of the plastics end-of-life fate based on OECD data of the year 2019 (OECD, 2022b). The "Figure 1.2." of this report gives the following flows for global plastics in 2019 in million tons (Mt):

- Plastics use: 460 Mt
- Plastics waste: 353 Mt
  - Collected for recycling: 55 Mt
    - Recycling residues: -22 Mt
  - Mismanaged: 79 Mt
    - Aquatic leakage: 9 Mt
    - Terrestrial leakage: 13 Mt
    - Open-pit burning: 26 Mt
    - Dumpsites: 34 Mt
  - Landfilled: 174 Mt
  - Incinerated: 67 Mt

A tiny percentage of the difference between use and waste is leaked to the environment, but the core of it is accumulated in the technosphere. We do not know the future fate of this accumulated plastic, but here we take the assumption that it will follow the same pattern as waste. The Table S3 below gives the possible end-of-life fates, their relative shares and their representative Ecoinvent activity.

Table S3: Fate, relative share of plastic waste (managed and unmanaged disposal) and corresponding representative Ecoinvent activity and geography.

Fate	*Share*	*Ecoinvent activity*
Recycling (net)	10%	Assuming no GHG footprint (cutoff)
Aquatic leakage	2%	Assuming no GHG footprint (no data)
Terrestrial leakage	4%	Assuming no GHG footprint (no data)
Open-pit burning	7%	treatment of waste plastic, mixture, open burning (GLO)
Dumpsites	10%	treatment of waste plastic, mixture, unsanitary landfill, dry infiltration class (100mm) (GLO)
Landfilled	49%	treatment of waste plastic, mixture, sanitary landfill (RoW)
Incinerated	19%	treatment of waste plastic, mixture, municipal incineration (RoW)

Considering that fossil-derived molecules end-of-life GHG emissions are directly related to the amount of carbon they contain, we scaled the different Ecoinvent activities based on the carbon mass fraction rather than of the total mass. The amount of carbon (C) per mass of each of the molecules was calculated based on their molecular formula and the atomic mass (see the SM Step 3 Excel file). The representative product of the Ecoinvent activities listed in Table S3 is "waste plastic, mixture", with a carbon content of 63.4% of which 100% is fossil, as of version 3.12.

The main limitations of our approach are notably the lack of representativeness of each specific product and the assumption of similar end of life for accumulated stocks. In addition, the illustrative demonstration focuses on petrochemicals, thus we considered that the selected product are 100% fossil-derived molecules. This implies that there is no biogenic or captured carbon embedded in the in them, which would otherwise have drastically different GWP contribution. Future work may attempt to expand the end-of-life scenarios to improve region-product-use specificity. As such, these datasets could potentially be used to collect data:

- OECD (2024a) "Plastic Waste - Estimations from 1990 to 2019." Available at: https://data-explorer.oecd.org/vis?pg=0&bp=true&snb=113&isAvailabilityDisabled=false&page=0&tm=Plastic%20Waste&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PW%40DF_PW&df[ag]=OECD.ENV.EEI&df[vs]=1.0 (Accessed: August 19, 2026).
- OECD (2024b) "Plastic Waste in 2019." Available at: https://data-explorer.oecd.org/vis?pg=0&bp=true&snb=113&isAvailabilityDisabled=false&page=0&tm=Plastic%20Waste&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PW_2019%40DF_PW_2019&df[ag]=OECD.ENV.EEI&df[vs]=1.0 (Accessed: August 19, 2026).
- OECD (2024c) "Plastics Use in 2019." Available at: https://data-explorer.oecd.org/vis?fs[0]=Topic%2C1%7CEnvironment%20and%20climate%20change%23ENV%23%7CPlastics%23ENV_PLS%23&pg=0&fc=Topic&bp=true&snb=11&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PU_2019%40DF_PU_2019&df[ag]=OECD.ENV.EEI&df[vs]=1.0&dq=...TOTAL&pd=2019%2C&to[TIME_PERIOD]=false&vw=ov (Accessed: April 17, 2025).

4.5.Step 4 data and code

We provide a Python notebook to automatize the PRODCOM→HS22 matching procedure. Given an input CSV file containing PRODCOM codes, the code gives the corresponding HS codes. This drastically reduces manual work, errors and favours reproducibility. It is available in the Git repository at this location: "Python_notebooks/Convert PRODCOM codes to Harmonized System (HS) 2022 codes.ipynb".

The SM Excel file "Step 3 - PRODCOM results" (mmc3.xlsx) contains the mapped Ecoinvent activities, the geography shares of each supported Ecoinvent database version, the carbon content. In addition, there are specific sheets, one for the end-of-life fates and one for the calculations to split the aggregated polyamides and isocyanates into specific products (i.e., two major products for both cases) with a matching Ecoinvent activity.

The trade data and Ecoinvent geographies discovery code is located in this Python notebook: "Python_notebooks/Get product importers from BACI and Ecoinvent geographies.ipynb".

Once the list of selected hotspot products is equipped with Ecoinvent geographies and respective their share, their carbon content specified, they are ready for footprint calculations. Unit footprint, substance contribution and end-of-life footprints are calculated within this Python notebook: "Python_notebooks/Calculate hotspot products footprints and contributions.ipynb".

Once the footprint results have been calculated, there is a specific Python notebook dedicated to generate the main paper's Figure 2 and it is available here: "Python_notebooks/Generate main paper hotspot figure 2.ipynb"

The SM Excel file "Step 4 - Figure 2 data - Hotspot products" (mmc4.xlsx) contains the results behind the main paper's Figure 2 with additional products. There is also a sheet containing the calculations required to estimate substance contributions to the GHG footprint of PA-6 and PA-6,6.

5. Sensitivity analysis (steps 2 to 4)

This section provides some insights gained through the different sensitivity analysis that were run on the data. All of the code necessary to generate the required data for the analysis are included in the documented Python notebooks available online in the Git repositories. Some of the input and intermediate data used for the analysis are not provided due to some constraints (see "

Git repositor" section above).

5.1. EXIOBASE: product sector's baseline footprint

In the main article, EXIOBASE is used to calculate the results for France and for the year 2022. As part of the sensitivity analysis, we calculated the GHG footprint results with different scopes. The two products analysed (i.e., "Plastics, basic" and "Chemicals nec") remain constant, the same for the EXIOBASE version (i.e., v3.10.2). We tested these new configurations: EU27 instead of France for the region, and 2019 instead of 2022 for the year. Indeed, EU27 strongly connected to step 3 which analyses PRODCOM data at this regional level. When it comes to the time series, 2019 was tested against 2022. It is a close year while not being drastically too far from the current situation. Taking something in between could also lack representativeness, due to the disruptions caused by the COVID-19 pandemic on the industry. For consistency in our analysis, EU27 includes the same member countries for both 2019 and 2022, even though the "EU27" was only created in 2020. Results were calculated in terms of GHG footprint for the given product, region and year. They include the local production and imported products. We also report the regions' consumption GHG footprint (consumption-based accounts, CBA) and its direct emissions for the same year (production-based accounts, PBA) for comparison (Wood *et al.*, 2020; Stadler, 2021). Main results are presented in the **Error! Reference source not found.**. We can conclude that the relative importance of the two chosen products is not sensitive when switching to other relevant year or region. The main result from step 1 remains unchanged, plastics and chemicals are key sectors over which France (or the EU27) have some control.

Table S4: Sensitivity results for EXIOBASE, values in 220.3 Mt CO_2-eq. Percentage of PBA and CBA provided in brackets should be interpreted with caution: The product footprints are not necessarily part of PBA and CBA, they don't follow the same accounting method while both being expressed in the same unit of measurement. Instead, they are given as a relative share to give an idea of the significance of values.

	France (FR) 2019	*France (FR) 2022*	*European Union (EU27) 2019*	*European Union (EU27) 2022*
Production-based accounts (PBA)	582	592	4,450	4,440
Consumption-based accounts (CBA)	416	390	3,560	3,360
Plastics, basic	21.3 (~5.1% of PBA, ~3.7% of CBA)	21.7 (~5.6% of PBA, ~3.7% of CBA)	158.2 (~4.4% of PBA, ~3.7% of CBA)	220.3 (~6.6% of PBA, ~5% of CBA)
Chemicals nec	23.8 (~5.7% of PBA, ~4.1% of CBA)	22.2 (~5.7% of PBA, ~3.8% of CBA)	224.4 (~6.3% of PBA, ~5.0% of CBA)	241.4 (~7.2% of PBA, ~5.4% of CBA)

5.2. PRODCOM: sector's individual product quantities breakdown

The sensitivity analysis performed over PRODCOM data follows a similar procedure than that of EXIOBASE. Data were compared between the 2019 and 2022 time series, and between France and the EU27. Although, the main analysis was done over EU27 data due to frequent data gaps at France scale mentioned in the section 3.1.

The main results show that when comparing PRODCOM data for France and the EU27, we see broadly a similar pattern for the largest products. However, we note a lot of data gaps (i.e., "secrets") undermining the reliability of the information. Nonetheless, several products appear potentially underestimated when take the EU27 as a proxy for France. These are listed below:

- Polyvinyl chloride, not mixed with any other substances, in primary forms
- Linear polyethylene having a specific gravity < 0,94, in primary forms
- Buta-1,3-diene and isoprene
- Polystyrene, in primary forms (excluding expansible polystyrene)
- Vinyl chloride (chloroethylene)
- Hexamethylenediamine and its salts; ethylenediamine and its salts

On the other hand, a few products seem to be potentially overestimated with EU27 data:

- Methanol (methyl alcohol)
- 1,2-Dichloroethane (ethylene dichloride)
- p-Xylene

We also compared 2022 data with 2019 data for the EU27. We compare d the two series by deriving the relative share of each product to the sum of products quantity, after filtering out products. The resulting Pearson correlation coefficient between both series is $R^2 = 0.994$, implying a good correlation in relative share of each product when compared across different time series. Therefore, the chosen year here has minimal influence over the relative share of products, which is a key property for the following calculation of the footprint and prioritization of hotspots.

5.3. Ecoinvent: individual product footprints

We have performed a sensitivity analysis for each dataset across the available geographies (or location). Indeed, variations of the regional share calculated with PRODCOM and BACI trade data can directly shift the resulting Ecoinvent geography shares. For each product passed to the footprint calculation we have corresponding Ecoinvent activity, we extracted all available location for this activity and calculated the related GHG footprint using GWP_{100} from IPCC 2021 method. Then we proceeded to calculate the coefficient of variation (C.V.) of the footprint across available locations for each dataset and for both versions of Ecoinvent. The C.V. is the relative standard deviation to the average of the series and it allows us to compare all products. It was calculated solely for products with multiple regions available to avoid false accuracy. The results are summarized in the Table S5. The main result is the significant variability in the footprints of different Ecoinvent dataset locations. On average, the C.V. is equal to 15% for Ecoinvent 3.11 and 17% for version 3.12. This translates into the standard deviation being higher than the average footprint by 15% and 17% respectively, on average.

Table S5: Statistical analysis results of the coefficient of variation (C.V.) of the unit GHG footprint of the selected products. Only products with at least two different locations available in the Ecoinvent database are included. Results are given per Ecoinvent version, which differ in terms of the location availability. Unit: % (σ in kg CO_2-eq / μ in kg CO_2-eq).

	Ecoinvent 3.11 cutoff C.V.	*Ecoinvent 3.12 cutoff C.V.*
Average	15.0%	17.2%
Median	10.3%	10.7%
Min	0.4%	0.2%
Max	102.7%	99.6%
Standard deviation	0.210	0.217

We have also tested the influence of the choice of the Ecoinvent version on the footprint calculation. However, considering that the versions do not have the same regions being available for a same product this can only be performed for products where Ecoinvent provides the same locations in both versions. For this, we started from the regions available for the selected products in Ecoinvent 3.11, Rest-of-World (RoW) is excluded because this Ecoinvent geography's extent is not stable across datasets and versions. We then proceeded to compare these with the values available in Ecoinvent 3.12. There were 50 Ecoinvent datasets (product with specific location) with a match in both databases and considering the aforementioned exclusion. We could determine an average decrease in the footprint of 2% from Ecoinvent v3.11 to v3.12, with a standard deviation of 8% and a median of less than 0.3%. The difference between the two consecutive versions is therefore contained, leading to low apparent sensitivity.

Finally, we note that moving from Ecoinvent 3.11 to 3.12 allows more recent data and at the same time more precise regions, thereby making it possible to represent each product's regional share more closely to the reality. Our results show that the results changed minimally. The final unit footprint without the end-of-life, not being influenced by Ecoinvent, is on average 1% lower in 3.12 compared to 3.11, the standard deviation is around 7% and the median 0%. This implies small changes on average, which tend to cancel out across the different products. The most critical change concerns polypropylene, which has reduced by 23%. This has no implication in the relative ranking of polypropylene, which remains second in terms of its total footprint including end-of-life estimation for any of the two Ecoinvent version used. However, we note a swap between benzene and PET position in the final ranking, when comparing analysis conducted with Ecoinvent v3.11 and v3.12. This is due to their relatively close footprint (i.e., 18.5 Mt CO_2-eq for benzene and 18.2 for PET in v3.12), hence the tiny reduction in the production footprint of benzene and tiny increase in the case of PET resulted in visible change in their relative position. This is the only change in the top 15 hotspot products ranking when comparing the two versions, the next absolute difference in the ranking position is seen after the 23rd hotspot product. As a result, we see minimal influence of the database version over the key result of this method being is the ranking hotspots in terms of their total GHG footprint.

5.4.Sensitivity analysis data and code

The EXIOBASE related sensitivity uses the data available in the Excel file "Step 2 - EXIOBASE results" (mmc2.xlsx) in the SM.

The results of section “5.2 PRODCOM: sector’s individual product quantities breakdown” are provided in the Excel file “Step 3 - PRODCOM results” (mmc3.xlsx) available in the SM, and data can be compared in the sheet “Filtered PRODCOM data” by looking at the table’s following columns:

- “PRODQNT + IMPQNT - EU27 - 2019”
- “PRODQNT + IMPQNT - EU27 - 2022”
- “PRODQNT + IMPQNT - FR - 2019”
- “PRODQNT + IMPQNT - FR - 2022”

Each row represents a product and the columns show the available quantity expressed in kilogram of product, calculated as the sum of imported and locally produced quantity for the product in the given region and year. The accompanying column right next each tells if there are some data gaps for the product.

The code to calculate the footprint of the different Ecoinvent geographies for the sensitivity analysis is in this Python notebook: “Python_notebooks/Calculate Ecoinvent unit product footprint for sensitivity analysis.ipynb”

The SM Excel “Step 4 - Figure 2 data - Hotspot products” (mmc4.xlsx) contains the sensitivity calculations of Ecoinvent geographies and the overall sensitivity of the ranking results depending on the Ecoinvent version used.

6. Evaluating the process maturity (step 7)

Table S6: Process technological maturity assessment rules for simplified 4-stage TRL.

Maturity	*Description*	*Typical information leading to rating*
Fully mature	Processes that are representative of the market. It can be the current petrochemical process used.	The process is introduced as being the conventional method to date or one of the most used.
High	The process has reached an industrial scale, even if the market is small.	Industrial, commercialized, mature, >1000 t/year.
Medium	The process has passed the lab scale and is now tested with industrial partners while not having found an economically viable state.	Demonstration, pilot scale, >1 t/year.
Low	The process is still at lab scale or conceptual.	Lab scale, bench scale, research state.

7. Bio-based pathways graph database (steps 5 to 7)

7.1. Kumu graph database link

Available at https://kumu.io/nicolas-lienart/screen-lca-bio-based-pathways-database (raw JSON and Excel database files are available in the Git repository's "Kumu_data/").

The database may receive future updates. To view the data as of the published version of the article, open the following map: "Published version (v1)".

7.2. References used in the graph database

Eckert, M., Fleischmann, G., Jira, R., Bolt, H. M., & Golka, K. (2006). Acetaldehyde. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a01_031.pub2

Campbell, M. L. (2011). Cyclohexane. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a08_209.pub2

Bazzanella, A., & Ausfelder, F. (2017). Low carbon energy and feedstock for the European chemical industry: Technology Study. DECHEMA, Gesellschaft für Chemische Technik und Biotechnologie eV.

Olabisi, O., & Adewale, K. (Eds.). (2016). Handbook of thermoplastics (Second edition). CRC Press, Taylor & Francis Group.

Vogt, E. T. C., & Weckhuysen, B. M. (2024). The refinery of the future. Nature, 629(8011), 295–306. https://doi.org/10.1038/s41586-024-07322-2

Le Berre, C., Serp, P., Kalck, P., & Torrence, G. P. (2014). Acetic Acid. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–34). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a01_045.pub3

Leaf by Lesaffre. (2024). White Paper on the Valorization of Cellulosic Sugars. https://content.leaf-lesaffre.com/white-paper-cellulosic-sugars-valorization

Held, H., Rengstl, A., & Mayer, D. (2000). Acetic Anhydride and Mixed Fatty Acid Anhydrides. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a01_065

Minx, J. C., Lamb, W. F., Callaghan, M. W., Fuss, S., Hilaire, J., Creutzig, F., Amann, T., Beringer, T., de Oliveira Garcia, W., Hartmann, J., Khanna, T., Lenzi, D., Luderer, G., Nemet, G. F., Rogelj, J., Smith, P., Vicente Vicente, J. L., Wilcox, J., & del Mar Zamora Dominguez, M. (2018). Negative emissions—Part 1: Research landscape and synthesis. Environmental Research Letters, 13(6), 063001. https://doi.org/10.1088/1748-9326/aabf9b

Srivastava, R. K., Sarangi, P. K., Bhatia, L., Singh, A. K., & Shadangi, K. P. (2022). Conversion of methane to methanol: technologies and future challenges. Biomass Conversion and Biorefinery, 12(5), 1851–1875. https://doi.org/10.1007/s13399-021-01872-5

de Jong, E., Higson, A., Walsh, P., & Wellisch, M. (2012). Product developments in the bio-based chemicals arena. Biofuels, Bioproducts and Biorefining, 6(6), 606–624. https://doi.org/10.1002/bbb.1360

Rosenboom, J.-G., Langer, R., & Traverso, G. (2022). Bioplastics for a circular economy. Nature Reviews Materials, 7(2), 117–137. https://doi.org/10.1038/s41578-021-00407-8

Lang, M., & Li, H. (2022). Sustainable Routes for the Synthesis of Renewable Adipic Acid from Biomass Derivatives. ChemSusChem, 15(1), e202101531. https://doi.org/10.1002/cssc.202101531

Ziegler-Skylakakis, K., Fabri, J., Graeser, U., & Simo, T. A. (2019). Xylenes. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–20). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a28_433.pub2

Clark, J. H., Farmer, T. J., Hunt, A. J., & Sherwood, J. (2015). Opportunities for Bio-Based Solvents Created as Petrochemical and Fuel Products Transition towards Renewable Resources. International Journal of Molecular Sciences, 16(8), 17101–17159. https://doi.org/10.3390/ijms160817101

Hietala, J., Vuori, A., Johnsson, P., Pollari, I., Reutemann, W., & Kieczka, H. (2016). Formic Acid. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–22). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a12_013.pub3

Herzog, B. D., & Smiley, R. A. (2012). Hexamethylenediamine. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a12_629.pub2

Ziegler-Sylakakis, K., Fabri, J., Graeser, U., & Simo, T. A. (2019). Toluene. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–12). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a27_147.pub3

Kumar, L., Asthana, S., Laxman Newalkar, B., & Kishore Pant, K. (2024). Selective toluene methylation to p-xylene: current status & future perspective. Catalysis Reviews, 66(3), 820–862. https://doi.org/10.1080/01614940.2022.2097641

Hesse, W., & Lang, J. (2011). Phenolic Resins. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a19_371.pub2

Siegenthaler, K. O., Jung, P., Künkel, A., Lohmann, J., Loos, R., Neumann, P., Parlings, M., Porc, O., Schröder, H., Van Berkel, J., Xi, L., & Yamamoto, M. (2022). Polymers, Biobased Structural. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–18). https://doi.org/10.1002/14356007.v04_v01

Grasselli, R. K., & Trifirò, F. (2016). Acrylonitrile from Biomass: Still Far from Being a Sustainable Process. Topics in Catalysis, 59(17), 1651–1658. https://doi.org/10.1007/s11244-016-0679-7

Chen, Y., & Nielsen, J. (2013). Advances in metabolic pathway and strain engineering paving the way for sustainable production of chemical building blocks. Current Opinion in Biotechnology, 24(6), 965–972. https://doi.org/10.1016/j.copbio.2013.03.008

Buznitsky, K., Verma, S., & Nitzsche, M. P. (2024). Decarbonizing industry: Policy approaches to eliminate hard-to-abate emissions. MIT Science Policy Review. https://sciencepolicyreview.pubpub.org/pub/3v3e6ed7/release/2

Wang, Z., Ganewatta, M. S., & Tang, C. (2020). Sustainable polymers from biomass: Bridging chemistry with materials and processing. Progress in Polymer Science, 101, 101197. https://doi.org/10.1016/j.progpolymsci.2019.101197

Mori, R. (2023). Replacing all petroleum-based chemical products with natural biomass-based chemical products: a tutorial review. RSC Sustainability, 1(2), 179–212. https://doi.org/10.1039/D2SU00014H

Han, J. (2017). Biorenewable Strategy for Catalytic ε-Caprolactam Production Using Cellulose- and Hemicellulose-Derived γ-Valerolactone. ACS Sustainable Chemistry & Engineering, 5(2), 1892–1898. https://doi.org/10.1021/acssuschemeng.6b02616

Maki, T., & Takeda, K. (2000). Benzoic Acid and Derivatives. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a03_555

FranceAgriMer. (2024). Évolution du positionnement de la France sur le marché international des biocarburants. https://www.franceagrimer.fr/Autres-filieres/Actualites/Evolution-du-positionnement-de-la-France-sur-le-marche-international-des-biocarburants

Główka, M., & Krawczyk, T. (2023). New Trends and Perspectives in Production of 1,2-Propanediol. ACS Sustainable Chemistry & Engineering, 11(19), 7274–7287. https://doi.org/10.1021/acssuschemeng.3c01018

Winter, B., Meys, R., & Bardow, A. (2021). Towards aromatics from biomass: Prospective Life Cycle Assessment of bio-based aniline. Journal of Cleaner Production, 290, 125818. https://doi.org/10.1016/j.jclepro.2021.125818

Slak, J., Pomeroy, B., Kostyniuk, A., Grilc, M., & Likozar, B. (2022). A review of bio-refining process intensification in catalytic conversion reactions, separations and purifications of

hydroxymethylfurfural (HMF) and furfural. Chemical Engineering Journal, 429, 132325. https://doi.org/10.1016/j.cej.2021.132325

EC. (2019). Top 20 innovative bio-based products: task 3 of "Study on Support to R&I Policy in the Area of Bio based Products and Services." Publications Office of the European Union. https://data.europa.eu/doi/10.2777/85805

Klabunde, J., Bischoff, C., & Papa, A. J. (2018). Propanols. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–14). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a22_173.pub3

Marchiori, P., Ferreira, V. C., Fröner-Lacerda, L. R. R., Castro, L. E. N., Sillero, L., Zanin, H. G., & Forster-Carneiro, T. (2025). Global trends and challenges in biogas purification and CO2 capture for renewable energy and climate mitigation. Biofuels, Bioproducts and Biorefining, 19(6), 2474–2494. https://doi.org/10.1002/bbb.2805

Yahyazadeh, A., Nanda, S., & Dalai, A. K. (2024). A Critical Review of the Sustainable Production and Application of Methanol as a Biochemical and Bioenergy Carrier. Reactions, 5(1), 1–19. https://doi.org/10.3390/reactions5010001

Dye, R. F. (2001). Ethylene glycols technology. Korean Journal of Chemical Engineering, 18(5), 571–579. https://doi.org/10.1007/BF02706370

Diem, H., Matthias, G., & Wagner, R. A. (2010). Amino Resins. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a02_115.pub2

Richter, F., Karafilidis, C., & Darmandeh, H. (2024). Isocyanates, Organic. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–23). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a14_611.pub2

Gubbels, E., Heitz, T., Yamamoto, M., Chilekar, V., Zarbakhsh, S., Gepraegs, M., Köpnick, H., Schmidt, M., Brügging, W., Rüter, J., & Kaminsky, W. (2018). Polyesters. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–30). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a21_227.pub2

Thaore, V., Chadwick, D., & Shah, N. (2018). Sustainable production of chemical intermediates for nylon manufacture: A techno-economic analysis for renewable production of caprolactone. Chemical Engineering Research and Design, 135, 140–152. https://doi.org/10.1016/j.cherd.2018.05.026

Tinge, J., Groothaert, M., op het Veld, H., Ritz, J., Fuchs, H., Kieczka, H., & Moran, W. C. (2018). Caprolactam. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–31). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a05_031.pub3

Tran, H.-V., Dang, T. T., Nguyen, N. H., Tran, H. T., Nguyen, D. T., Do, D. V., Le, T. S., Ngo, T. H., Late, Y. K. E., Amaniampong, P. N., Fletcher, E., Hung, T. Q., Cheng, Y., Nguyen, T.-K., Tran, T. S., Zhang, J., An, H., Nguyen, N.-T., & Trinh, Q. T. (2024). Methanol Activation: Strategies for Utilization of Methanol as C1 Building Block in Sustainable Organic Synthesis. ChemSusChem, n/a(n/a), e202401974. https://doi.org/10.1002/cssc.202401974

Li, Y., Luo, X., & Hu, S. (2015). Bio-based Polyols and Polyurethanes. Springer International Publishing. https://doi.org/10.1007/978-3-319-21539-6

De Vos, L., Van de Voorde, B., Van Daele, L., Dubruel, P., & Van Vlierberghe, S. (2021). Poly(alkylene terephthalate)s: From current developments in synthetic strategies towards applications. European Polymer Journal, 161, 110840. https://doi.org/10.1016/j.eurpolymj.2021.110840

Teong, S. P., Yi, G., & Zhang, Y. (2014). Hydroxymethylfurfural production from bioresources: past, present and future. Green Chemistry, 16(4), 2015–2026. https://doi.org/10.1039/C3GC42018C

Fischer, I., Schmitt, W. F., Porth, H.-C., Allsopp, M. W., & Vianello, G. (2014). Poly(Vinyl Chloride). In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–30). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a21_717.pub2

Broda, M., Yelle, D. J., & Serwańska, K. (2022). Bioethanol Production from Lignocellulosic Biomass—Challenges and Solutions. Molecules, 27(24), 8717. https://doi.org/10.3390/molecules27248717

Löser, C., Urit, T., & Bley, T. (2014). Perspectives for the biotechnological production of ethyl acetate by yeasts. Applied Microbiology and Biotechnology, 98(12), 5397–5415. https://doi.org/10.1007/s00253-014-5765-9

Dros, A. B., Larue, O., Reimond, A., Campo, F. D., & Pera-Titus, M. (2015). Hexamethylenediamine (HMDA) from fossil- vs. bio-based routes: an economic and life cycle assessment comparative study. Green Chemistry, 17(10), 4760–4772. https://doi.org/10.1039/C5GC01549A

Franz, A. W., Kronemayer, H., Pfeiffer, D., Pilz, R. D., Reuss, G., Disteldorf, W., Gamer, A. O., & Hilt, A. (2016). Formaldehyde. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–34). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a11_619.pub2

Sattler, H., & Schweizer, M. (2011). Fibers, 5. Polyester Fibers. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.o10_o01

Weissermel, K., & Arpe, H. (2003). Industrial Organic Chemistry (1st ed.). Wiley. https://doi.org/10.1002/9783527619191

Lee, K., Jing, Y., Wang, Y., & Yan, N. (2022). A unified view on catalytic conversion of biomass and waste plastics. Nature Reviews Chemistry, 6(9), 635–652. https://doi.org/10.1038/s41570-022-00411-8

Molenveld, K., & Bos, H. (2020). Biobased Plastics 2020. Wageningen Food & Biobased Research. https://doi.org/10.18174/534587

Wegelin, S., & Meier, M. A. R. (2024). Bio-based aromatics for chemicals and materials: Advances in renewable drop-in and functional alternatives. Current Opinion in Green and Sustainable Chemistry, 47, 100931. https://doi.org/10.1016/j.cogsc.2024.100931

Singh, A., Gupta, A., Rakesh N., Shivapuji, A. M., & Dasappa, S. (2022). Syngas generation for methanol synthesis: oxy-steam gasification route using agro-residue as fuel. Biomass Conversion and Biorefinery, 12(5), 1803–1818. https://doi.org/10.1007/s13399-021-02128-y

Xu, C., Paone, E., Rodríguez-Padrón, D., Luque, R., & Mauriello, F. (2020). Recent catalytic routes for the preparation and the upgrading of biomass derived furfural and 5-hydroxymethylfurfural. Chemical Society Reviews, 49(13), 4273–4306. https://doi.org/10.1039/D0CS00041H

IFPEN. (2019, September 26). Michelin, IFP Energies nouvelles, and Axens give a new dimension to the BioButterfly project. IFPEN. https://www.ifpenergiesnouvelles.com/article/michelin-ifp-energies-nouvelles-and-axens-give-new-dimension-biobutterfly-project

Schmidt, O., Gambhir, A., Staffell, I., Hawkes, A., Nelson, J., & Few, S. (2017). Future cost and performance of water electrolysis: An expert elicitation study. International Journal of Hydrogen Energy, 42(52), 30470–30492. https://doi.org/10.1016/j.ijhydene.2017.10.045

Rosenboom, J.-G., Langer, R., & Traverso, G. (2022). Bioplastics for a circular economy. Nature Reviews Materials, 7(2), 117–137. https://doi.org/10.1038/s41578-021-00407-8

Lim, D., Lee, B., Lee, H., Byun, M., & Lim, H. (2022). Projected cost analysis of hybrid methanol production from tri-reforming of methane integrated with various water electrolysis systems: Technical and economic assessment. Renewable and Sustainable Energy Reviews, 155, 111876. https://doi.org/10.1016/j.rser.2021.111876

Liptow, C., Tillman, A.-M., & Janssen, M. (2015). Life cycle assessment of biomass-based ethylene production in Sweden — is gasification or fermentation the environmentally preferable route? The International Journal of Life Cycle Assessment, 20(5), 632–644. https://doi.org/10.1007/s11367-015-0855-1

Maul, J., Frushour, B. G., Kontoff, J. R., Eichenauer, H., Ott, K.-H., & Schade, C. (2007). Polystyrene and Styrene Copolymers. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a21_615.pub2

Toor, M., Kumar, S. S., Malyan, S. K., Bishnoi, N. R., Mathimani, T., Rajendran, K., & Pugazhendhi, A. (2020). An overview on bioethanol production from lignocellulosic feedstocks. Chemosphere, 242, 125080. https://doi.org/10.1016/j.chemosphere.2019.125080

Periyasamy, S., Karthik, V., Senthil Kumar, P., Isabel, J. B., Temesgen, T., Hunegnaw, B. M., Melese, B. B., Mohamed, B. A., & Vo, D.-V. N. (2022). Chemical, physical and biological methods to convert lignocellulosic waste into value-added products. A review. Environmental Chemistry Letters, 20(2), 1129–1152. https://doi.org/10.1007/s10311-021-01374-w

Bienewald, F., Leibold, E., Tužina, P., & Roscher, G. (2019). Vinyl Esters. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–16). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a27_419.pub2

Akindoyo, J. O., Beg, M. D. H., Ghazali, S., Islam, M. R., Jeyaratnam, N., & Yuvaraj, A. R. (2016). Polyurethane types, synthesis and applications – a review. RSC Advances, 6(115), 114453–114482. https://doi.org/10.1039/C6RA14525F

Zacharopoulou, V., & Lemonidou, A. A. (2018). Olefins from Biomass Intermediates: A Review. Catalysts, 8(1), 2. https://doi.org/10.3390/catal8010002

Choi, S., Song, C. W., Shin, J. H., & Lee, S. Y. (2015). Biorefineries for the production of top building block chemicals and their derivatives. METABOLIC ENGINEERING, 28, 223–239. https://doi.org/10.1016/j.ymben.2014.12.007

Walters, P., Cadogan, D. F., & Howick, C. J. (2020). Plasticizers. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–27). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a20_439.pub2

Sahota, S., Lombardi, L., Singh, D., O'Donoghue, C., Geoghegan, C., Rova, U., Christakopoulos, P., Matsakas, L., & Sarkar, O. (2025). CO2 from biogas: valorisation, economic and environmental impacts in circular carbon systems. Reviews in Environmental Science and Bio/Technology, 25(1), 6. https://doi.org/10.1007/s11157-025-09755-1

Albrecht, K., Stickler, M., & Rhein, T. (2013). Polymethacrylates. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a21_473.pub2

Deka, T. J., Osman, A. I., Baruah, D. C., & Rooney, D. W. (2022). Methanol fuel production, utilization, and techno-economy: a review. Environmental Chemistry Letters, 20(6), 3525–3554. https://doi.org/10.1007/s10311-022-01485-y

Gail, E., Gos, S., Kulzer, R., Lorösch, J., Rubo, A., Sauer, M., Kellens, R., Reddy, J., Steier, N., & Hasenpusch, W. (2011). Cyano Compounds, Inorganic. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a08_159.pub3

Soleimani, S., & Lehner, M. (2022). Tri-Reforming of Methane: Thermodynamics, Operating Conditions, Reactor Technology and Efficiency Evaluation—A Review. Energies, 15(19), 7159. https://doi.org/10.3390/en15197159

Beerthuis, R., Rothenberg, G., & Shiju, N. R. (2015). Catalytic routes towards acrylic acid, adipic acid and ε-caprolactam starting from biorenewables. Green Chemistry, 17(3), 1341–1361. https://doi.org/10.1039/C4GC02076F

Zhang, Z., Delcroix, B., Rezazgui, O., & Mangin, P. (2021). Simulation and techno-economic assessment of bio-methanol production from pine biomass, biochar and pyrolysis oil. Sustainable Energy Technologies and Assessments, 44, 101002. https://doi.org/10.1016/j.seta.2021.101002

Crews, G. M., Ripperger, W., Kersebohm, D. B., Güthner, T., & Mertschenk, B. (2006). Melamine and Guanamines. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a16_171.pub2

Welch, V. A., Fallon, K. J., & Gelbke, H.-P. (2005). Ethylbenzene. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a10_035.pub2

Ceresco, B4C, & CoDEM. (2025). Etude de marché des produits biosourcés. ADEME. https://librairie.ademe.fr/agriculture-alimentation-foret-bioeconomie/8040-etude-de-marche-des-produits-biosources.html

Naser, A., Deiab, I., & Darras, B. M. (2021). Poly(lactic acid) (PLA) and polyhydroxyalkanoates (PHAs), green alternatives to petroleum-based plastics: a review. RSC Advances, 11(28), 17151–17196. https://doi.org/10.1039/D1RA02390J

Davidson, M. G., Elgie, S., Parsons, S., & Young, T. J. (2021). Production of HMF, FDCA and their derived products: a review of life cycle assessment (LCA) and techno-economic analysis (TEA) studies. Green Chemistry, 23(9), 3154–3171. https://doi.org/10.1039/D1GC00721A

Haubs, M., Kurz, K., & Sextro, G. (2012). Polyoxymethylenes. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a21_591.pub2

Raff, D. K. (2013). Butanals. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a04_447.pub2

IEA. (2018). The Future of Petrochemicals. https://www.iea.org/reports/the-future-of-petrochemicals

Baer, H., Bergamo, M., Forlin, A., Pottenger, L. H., & Lindner, J. (2012). Propylene Oxide. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a22_239.pub3

López-Gómez, J. P., Pérez-Rivero, C., & Venus, J. (2020). Valorisation of solid biowastes: The lactic acid alternative. Process Biochemistry, 99, 222–235. https://doi.org/10.1016/j.procbio.2020.08.029

Wu, Y., Yang, J., Wu, G., Gao, W., Silva Lora, E. E., Isa, Y. M., Subramanian, K. A., Kozlov, A., Zhang, S., & Huang, Y. (2023). Benzene, Toluene, and Xylene (BTX) Production from Catalytic Fast Pyrolysis of Biomass: A Review. ACS Sustainable Chemistry & Engineering, 11(32), 11700–11718. https://doi.org/10.1021/acssuschemeng.3c01202

Tomás, R. A. F., Bordado, J. C. M., Gomes, J. F. P., & Sheehan, R. J. (2024). Terephthalic Acid, Dimethyl Terephthalate, and Isophthalic Acid. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–17). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a26_193.pub3

Wong, M. K., Lock, S. S. M., Chan, Y. H., Yeoh, S. J., & Tan, I. S. (2023). Towards sustainable production of bio-based ethylene glycol: Progress, perspective and challenges in catalytic conversion and purification. Chemical Engineering Journal, 468, 143699. https://doi.org/10.1016/j.cej.2023.143699

Khedr, M. S. F. (2023). Bio-based polyamide. Physical Sciences Reviews, 8(7), 827–847. https://doi.org/10.1515/psr-2020-0076

Devi, A., Singh, A., Bajar, S., Pant, D., & Din, Z. U. (2021). Ethanol from lignocellulosic biomass: An in-depth analysis of pre-treatment methods, fermentation approaches and detoxification processes. Journal of Environmental Chemical Engineering, 9(5), 105798. https://doi.org/10.1016/j.jece.2021.105798

Hahn, H.-D., Dämbkes, G., Rupprich, N., Bahl, H., & Frey, G. D. (2013). Butanols. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a04_463.pub3

Tse, T. J., Wiens, D. J., & Reaney, M. J. T. (2021). Production of Bioethanol—A Review of Factors Affecting Ethanol Yield. Fermentation, 7(4), 268. https://doi.org/10.3390/fermentation7040268

Vu, H. P., Nguyen, L. N., Vu, M. T., Johir, M. A. H., McLaughlan, R., & Nghiem, L. D. (2020). A comprehensive review on the framework to valorise lignocellulosic biomass as biorefinery feedstocks. Science of The Total Environment, 743, 140630. https://doi.org/10.1016/j.scitotenv.2020.140630

Sardon, H., Mecerreyes, D., Basterretxea, A., Avérous, L., & Jehanno, C. (2021). From Lab to Market: Current Strategies for the Production of Biobased Polyols. ACS Sustainable Chemistry & Engineering, 9(32), 10664–10677. https://doi.org/10.1021/acssuschemeng.1c02361

James, D. H., & Castor, W. M. (2011). Styrene. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a25_329.pub2

Kiefer, D., Merkel, M., Lilge, L., Henkel, M., & Hausmann, R. (2021). From Acetate to Bio-Based Products: Underexploited Potential for Industrial Biotechnology. Trends in Biotechnology, 39(4), 397–411. https://doi.org/10.1016/j.tibtech.2020.09.004

Gautam, P., Neha, Upadhyay, S. N., & Dubey, S. K. (2020). Bio-methanol as a renewable fuel from waste biomass: Current trends and future perspective. Fuel, 273, 117783. https://doi.org/10.1016/j.fuel.2020.117783

Rodríguez-Vallejo, D. F., Guillén-Gosálbez, G., & Chachuat, B. (2020). What Is the True Cost of Producing Propylene from Methanol? The Role of Externalities. ACS Sustainable Chemistry & Engineering, 8(8), 3072–3081. https://doi.org/10.1021/acssuschemeng.9b05516

Siracusa, V., & Blanco, I. (2020). Bio-Polyethylene (Bio-PE), Bio-Polypropylene (Bio-PP) and Bio-Poly(ethylene terephthalate) (Bio-PET): Recent Developments in Bio-Based Polymers Analogous to Petroleum-Derived Ones for Packaging and Engineering Applications. Polymers, 12(8), 1641. https://doi.org/10.3390/polym12081641

Hamelinck, C. N., Suurs, R. A. A., & Faaij, A. P. C. (2005). International bioenergy transport costs and energy balance. Biomass and Bioenergy, 29(2), 114–134. https://doi.org/10.1016/j.biombioe.2005.04.002

Häussinger, P., Lohmüller, R., & Watson, A. M. (2011). Hydrogen, 2. Production. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.o13_o03

Lorz, P. M., Towae, F. K., Enke, W., Jäckh, R., Bhargava, N., & Hillesheim, W. (2007). Phthalic Acid and Derivatives. In Ullmann's Encyclopedia of Industrial Chemistry. John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a20_181.pub2

Muñoz, I., & Weidema, B. P. (2024). Ethylene and propylene production from steam cracking in Europe: a consequential perspective. The International Journal of Life Cycle Assessment, 29(4), 745–758. https://doi.org/10.1007/s11367-024-02282-1

Periyasamy, S., Karthik, V., Senthil Kumar, P., Isabel, J. B., Temesgen, T., Hunegnaw, B. M., Melese, B. B., Mohamed, B. A., & Vo, D.-V. N. (2022). Chemical, physical and biological methods to convert lignocellulosic waste into value-added products. A review. Environmental Chemistry Letters, 20(2), 1129–1152. https://doi.org/10.1007/s10311-021-01374-w

IEA Bioenergy. (2020). Bio-based chemicals: a 2020 update. https://www.ieabioenergy.com/wp-content/uploads/2020/02/Bio-based-chemicals-a-2020-update-final-200213.pdf

Roland-Holst, D., Triolo, R., Heft-Neal, S., & Bayrami, B. (2013). Bioplastics in california. http://www.bearecon.com/portfolio-data/bioplastics-california/bioplastics-california-report.pdf

McKenna, R., & Nielsen, D. R. (2011). Styrene biosynthesis from glucose by engineered E. coli. Metabolic Engineering, 13(5), 544–554. https://doi.org/10.1016/j.ymben.2011.06.005

Zimmermann, H. (2013). Propene. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 00–00). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.a22_211.pub3

Straathof, A. J. J. (2014). Transformation of Biomass into Commodity Chemicals Using Enzymes or Cells. Chemical Reviews, 114(3), 1871–1908. https://doi.org/10.1021/cr400309c

Werpy, T., & Petersen, G. (2004). Top Value Added Chemicals from Biomass: Volume I -- Results of Screening for Potential Candidates from Sugars and Synthesis Gas (DOE/GO-102004-1992, 15008859; p. DOE/GO-102004-1992, 15008859). https://doi.org/10.2172/15008859

IFPEN. (2025). Life cycle assessment of e-/bio- methanol and e-/grey-/blue- ammonia for maritime transport. https://www.ifpenergiesnouvelles.fr/sites/ifpen.fr/files/inline-images/20250310_IFPEN_CMACGM_ok.pdf

Dos Santos, L. V., De Barros Grassi, M. C., Gallardo, J. C. M., Pirolla, R. A. S., Calderón, L. L., De Carvalho-Netto, O. V., Parreiras, L. S., Camargo, E. L. O., Drezza, A. L., Missawa, S. K., Teixeira, G. S., Lunardi, I., Bressiani, J., & Pereira, G. A. G. (2016). Second-Generation Ethanol: The Need is Becoming a Reality. Industrial Biotechnology, 12(1), 40–57. https://doi.org/10.1089/ind.2015.0017

Künkel, A., Becker, J., Börger, L., Hamprecht, J., Koltzenburg, S., Loos, R., Schick, M. B., Schlegel, K., Sinkel, C., Skupin, G., & Yamamoto, M. (2016). Polymers, Biodegradable. In Ullmann's Encyclopedia of Industrial Chemistry (pp. 1–29). John Wiley & Sons, Ltd. https://doi.org/10.1002/14356007.n21_n01.pub2

Kikuchi, Y. et al. (2017) "Greenhouse gas emissions and socioeconomic effects of biomass-derived products based on structural path and life cycle analyses: A case study of polyethylene and polypropylene in Japan," Journal of Cleaner Production, 167, pp. 289–305. Available at: https://doi.org/10.1016/j.jclepro.2017.08.179.

8. Main output data

This section summarizes this research's available output data, focusing on the main outputs. It is presented per file:

- "Step 2 - EXIOBASE results" (mmc2.xlsx)
    - GHG footprint of the 200 traded product categories representing the whole economy of France and of the EU27 and for the year 2019 and 2022. The data distinguishes the origin (i.e., local production or imported products) and the destination (i.e., industry or final demand).
    - The total footprint and total direct emissions of the same regions and years.
- "Step 3 - PRODCOM results" (mmc3.xlsx)
    - Local production, imports and exports quantities in physical unit (i.e., kilogram mostly) of basic organic chemicals and plastics in France and in the EU27 for the years 2019 and 2022. Filtering of the top products to remove inorganics.
    - Disaggregating of some products into their major components: polyamides (PA-6 and PA-6,6) and isocyanates (MDI and TDI).
    - Match top products (N=57, including disaggregated products) to representative Ecoinvent activity, HS22 code and carbon content.
    - Estimated Ecoinvent geography representativity of the top products for the EU27.
    - Estimated generic end-of-life fate shares of the top products.
- "Step 4 - Figure 2 data - Hotspot products" (mmc4.xlsx)
    - The estimated GHG footprint of the production of the top products in the EU27, derived from the different representative origins.
    - The upstream GHG footprint contribution of ethylene, propylene, methanol and BTX to the products' footprint.
    - GHG footprint estimated of the end of life of these products.
    - Estimated total footprint, excluding the use phase, considered irrelevant due to the scope of this study.
- "Graph database of hotspot products and their bio-based pathways"
    - A graph database (based nodes interconnected with edges and augmented with attributes) to represent the bio-based production pathways.
    - The 50 hotspot products are represented with some additional products (mostly intermediates) totaling to more than 100 products. More than 100 intermediate processes are documented in between, which interconnect product nodes through directed edges (arrows).
    - Each node, whether a product or a process, is annotated with some metadata, including processes' literature references (see section 7.2).

9. Bibliography (specific to this file)

Aguilar-Hernandez, G.A. (2025) "A Novel Framework to Measure Circularity Trade-offs and Synergies in the Global Context," *Journal of Circular Economy*, 3(1). Available at: https://doi.org/10.55845/YTGD9041.

EC (2022) *Commission Delegated Regulation (EU) 2023/137 of 10 October 2022 amending Regulation (EC) No 1893/2006 of the European Parliament and of the Council establishing the statistical classification of economic activities NACE Revision 2 (Text with EEA relevance)*, *OJ L*. Available at: http://data.europa.eu/eli/reg_del/2023/137/oj (Accessed: February 5, 2026).

Ecoinvent (2025) "ecoinvent v3.12." Available at: https://ecoinvent.org/ecoinvent-v3-12/.

Eurostat (2023a) *European business statistics user's manual for PRODCOM: 2023 edition.* 2023 edition. LU: Publications Office. Available at: https://data.europa.eu/doi/10.2785/39767 (Accessed: May 15, 2024).

Eurostat (2023b) "Statistical Classification of Economic Activities in the European Community, Rev. 2.1 (NACE Rev. 2.1)." Available at: https://showvoc.op.europa.eu/#/datasets/ESTAT_Statistical_Classification_of_Economic_Activities_in_the_European_Community_Rev._2.1._(NACE_2.1).

Eurostat (2025) *NACE Rev. 2.1 – Statistical classification of economic activities in the European Union*. Available at: https://doi.org/10.2785/155339.

Gaulier, G. and Zignago, S. (2010) *BACI: International trade database at the product-level. The 1994-2007 version*. Working Papers 2010–23. CEPII. Available at: https://www.cepii.fr/CEPII/en/publications/wp/abstract.asp?NoDoc=2726.

Geyer, R., Jambeck, J.R. and Law, K.L. (2017) "Production, use, and fate of all plastics ever made," *Science Advances*, 3(7), p. e1700782. Available at: https://doi.org/10.1126/sciadv.1700782.

Herzog, B. *et al.* (2020) "Polyamides," *Ullmann's Encyclopedia of Industrial Chemistry*. John Wiley & Sons, Ltd, pp. 1–47. Available at: https://doi.org/10.1002/14356007.a21_179.pub4.

Hong, J. and Xu, X. (2012) "Environmental impact assessment of caprolactam production – a case study in China," *Journal of Cleaner Production*, 27, pp. 103–108. Available at: https://doi.org/10.1016/j.jclepro.2011.12.037.

IEA (2018) *The Future of Petrochemicals*. Available at: https://www.iea.org/reports/the-future-of-petrochemicals.

Khedr, M.S.F. (2023) "Bio-based polyamide," *Physical Sciences Reviews*, 8(7), pp. 827–847. Available at: https://doi.org/10.1515/psr-2020-0076.

Li, Y. (2007) *Life Cycle Assessment of Chemical Processes and Products*. Available at: http://www.lib.ncsu.edu/resolver/1840.16/3805 (Accessed: February 2, 2026).

Meys, R. *et al.* (2021) "Achieving net-zero greenhouse gas emission plastics by a circular carbon economy," *Science*, 374(6563), pp. 71–76. Available at: https://doi.org/10.1126/science.abg9853.

Miller, R.E. and Blair, P.D. (2022) *Input-Output Analysis: Foundations and Extensions*. 3rd ed. Cambridge University Press. Available at: https://doi.org/10.1017/9781108676212.

OECD (2022a) *Global Plastics Outlook: Economic Drivers, Environmental Impacts and Policy Options*. OECD. Available at: https://doi.org/10.1787/de747aef-en.

OECD (2022b) *Global Plastics Outlook: Policy Scenarios to 2060*. OECD Publishing. Available at: https://doi.org/10.1787/aa1edf33-en.

Petrochemicals Europe (2019) *Petrochemicals make things happen*. Poster. Available at: https://www.petrochemistry.eu/wp-content/uploads/2021/05/Petrochemistry-FlowChart_2019MC_V13-13092019-withoutFolds.pdf.

Ryberg, M., Laurent, A. and Hauschild, M.Z. (2018) *Mapping of global plastic value chain and plastic losses to the environment: with a particular focus on marine environment*, *Mapping of global plastic value chain and plastic losses to the environment: with a particular focus on marine environment*. Report. United Nations Environment Programme. Available at: https://www.unep.org/resources/report/mapping-global-plastics-value-chain-and-plastics-losses-environment-particular.

de Souza, N.R.D. *et al.* (2024) "Challenges and opportunities toward a sustainable bio-based chemical sector in Europe," *WIREs Energy and Environment*, 13(4), p. e534. Available at: https://doi.org/10.1002/wene.534.

Stadler, K. (2021) "Pymrio – A Python Based Multi-Regional Input-Output Analysis Toolbox," *Journal of Open Research Software*, 9(1). Available at: https://doi.org/10.5334/jors.251.

Stadler, K. *et al.* (2026) "EXIOBASE 3.10.2." Zenodo. Available at: https://doi.org/10.5281/zenodo.20051562.

UNEP (2011) *Global guidance principles for Life Cycle Assessment (LCA) databases: a basis for greener processes and products*. United Nations Environment Programme. Available at: https://digitallibrary.un.org/record/785158.

Wiedmann, T. (2017) "On the decomposition of total impact multipliers in a supply and use framework," *Journal of Economic Structures*, 6(1), p. 11. Available at: https://doi.org/10.1186/s40008-017-0072-0.

Wood, R. *et al.* (2020) "The structure, drivers and policy implications of the European carbon footprint," *Climate Policy*, 20(sup1), pp. S39–S57. Available at: https://doi.org/10.1080/14693062.2019.1639489.